

Mandatory Problems

1. Diabetes Dataset – Missing Value Imputation & Encoding

Dataset Link:

[Pima Indians Diabetes Dataset](#)

Use the Pima Indians Diabetes Database dataset.

Tasks:

- Load the dataset using Pandas.
 - Check missing values using `isnull().sum()`.
 - Replace invalid 0 values in columns like Glucose, BloodPressure, and BMI with NaN.
 - Apply:
 - Necessary Imputation (Must be done with the proper one)
 - Create a categorical column:
 - Age_Group
 - Young (<30)
 - Adult (30–50)
 - Senior (>50)
 - Apply Encoding on Age_Group. Choose the proper encoding technique and explain why it is the best one.
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2. Heart Disease Dataset – One Hot Encoding

Dataset Link:

[Heart Disease UCI Dataset](#)

Use the Heart Disease UCI Dataset dataset.

Tasks:

- Load the dataset.
 - Identify categorical columns.
 - Apply Encoding on:
 - cp (chest pain type)
 - thal
 - Compare dataframe shape before and after encoding.
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3. Tips Dataset – Label Encoding

Dataset Link:

[Seaborn Tips Dataset](#)

Use the Tips Dataset dataset.

Tasks:

- Load dataset using Seaborn.
 - Apply Encoding on:
 - sex
 - smoker
 - Create a new feature:
 - tip_category
 - Low (<3)
 - Medium (3–6)
 - High (>6)
 - Encode tip_category.
 - Choose the proper encoding technique and explain why it is the best one.
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4. Car Price Dataset – Train/Test Encoding Pipeline

Dataset Link:

[Car Price Prediction Dataset](#)

Use the Car Price Prediction Dataset dataset.

Tasks:

- Split the dataset into train and test sets first.
 - Apply:
 - Ordinal Encoding on fuel type (Petrol, Diesel, CNG)
 - One Hot Encoding on transmission type
 - Handle unknown categories in test data properly.
 - Concatenate encoded columns with original dataframe.
 - Drop original categorical columns.
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Optional Problems

5. Heart Disease Dataset

Dataset Link:

[Heart Disease UCI Dataset](#)

Use the Heart Disease UCI Dataset dataset.

Tasks:

- Do preprocessing that includes:
 - Missing value imputation
 - Outlier detection using IQR
 - One Hot Encoding
 - Ordinal Encoding
 - Use ColumnTransformer
 - Transform both training and testing datasets properly.
 - Display final transformed feature matrix shape.
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6. Diabetes Dataset – Compare Encoding Techniques

Dataset Link:

[Pima Indians Diabetes Dataset](#)

Use the Pima Indians Diabetes Database dataset.

Tasks:

- Create a categorical feature from BMI:
 - Underweight
 - Normal
 - Overweight
 - Obese
 - Apply:
 - Label Encoding
 - Ordinal Encoding
 - One Hot Encoding
 - Write a short conclusion on which encoding is best for this feature.
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7. Hotel Booking Dataset – High Cardinality Encoding

Dataset Link:

[Hotel Booking Demand Dataset](#)

Use the Hotel Booking Demand Dataset dataset.

Tasks:

- Load dataset.
 - Apply:
 - One Hot Encoding on hotel
 - Label Encoding on booking status column
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8. Employee Attrition Dataset – Advanced Encoding Challenge

Dataset Link:

IBM HR Analytics Employee Attrition Dataset

Use the IBM HR Analytics Employee Attrition Dataset dataset.

Tasks:

- Split the dataset into train and test sets.
- Detects all categorical columns automatically.
- Apply:
 - Ordinal Encoding for education level
 - One Hot Encoding for department
 - Label Encoding for target column
- Handle unseen categories in test data.
- Create a reusable preprocessing function. (Optional)
- Display transformed train and test datasets.